

# Assemble Six Bolt, Washer, and Nut Stacks SOP

One episode builds six identical stacks, one at a time, and lines them up in a row. A stack is a bolt, a washer, a plate, and a nut. The bolt stands head down in the build block, the washer threads on first, the plate goes on next so the bolt passes up through its hole, the nut spins down onto the plate, and the wrench snugs it. The finished stack then leaves the block and stands in its position in the row.

The build order never changes: bolt, washer, plate, nut, wrench, row. Stacks are built in order, stack 1 through stack 6, and stack N goes to row position N.

Every turn in this episode is a quarter turn. The arms cannot turn further than that in one motion, so a nut is run down the thread and a wrench is pulled around with repeated quarter turn cycles, never with one long twist.

The table is set up in one of three ways. Only the parts tray moves; the build block and the six row positions are in the same place in all three. The nut dish and the wrench rest sit on the side opposite the tray.

- **Config L:** the parts tray is against the left edge; the nut dish and the wrench rest are on the right.
- **Config M:** the parts tray is at the back-center, against the back edge behind the build block; the nut dish and the wrench rest are on the right.
- **Config R:** the parts tray is against the right edge, behind the row; the nut dish and the wrench rest are on the left.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the parts tray starts against the left edge (**Config L**), at the back-center (**Config M**) or against the right edge (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Supply side:** the nut dish and the wrench rest sit together on the side opposite the tray — the right in Config L and M, the left in Config R.
- **Same-side rule:** the gripper on the tray's side takes every bolt, washer, and plate and steadies the plate — the left gripper in Config L and M, the right gripper in Config R. The gripper on the supply side takes every nut and the wrench. No arm reaches across the table, and nothing is passed hand to hand.

- **Fixed roles:** the build block stays fixed in the middle, the row stays along the front from the middle to the right, and the **right gripper** carries every finished stack to the row in all three configs.

## Setup

Complete both checklists before starting an episode.

### Cell configuration

- **Environment camera:** 900 mm.
- **cell\_type:** bimanual

### Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the parts tray in its config position, the build block in the middle, the nut dish and the wrench rest on the supply side, and all six row positions along the front.
3. Both arms are at home with grippers open.
4. The build block is fixed to the tabletop and does not move when pushed.
5. The block pocket holds a bolt head so the bolt cannot turn, and it is as deep as a bolt head, so a seated head sits flush with the block face and a washer lies flat across both.
6. The tabletop is clear of anything but the zones listed below.

### Materials checklist

1. The parts tray holds the bolt holder, the washer dish, and the plate row, and sits in the position for the episode's config:
  - **Config L:** against the left edge.
  - **Config M:** at the back-center, against the back edge behind the build block.
  - **Config R:** against the right edge, behind the row.
2. The bolt holder holds six identical bolts standing head down with their threaded ends up, one per hole, none touching.
3. The washer dish holds six identical washers lying flat, none stacked on another.
4. Six identical plates lie flat in a row on the tray, not touching, each with one hole through its center.
5. The nut dish sits beside the build block on the supply side (right in Config L and M, left in Config R) and holds six identical nuts lying flat, holes facing up, none stacked or touching.
6. Every bolt, washer, plate, and nut is clean, straight, and undamaged, and every nut runs down every bolt by hand.

7. The bolts are long enough that thread still shows above the nut when the nut is snug.
8. The wrench is a 12 point box end wrench that fits the nut flats. It lies flat in its rest beside the nut dish on the supply side with the handle clear, so a gripper can take it from above.
9. The six row positions are marked along the front of the tabletop, numbered 1 to 6 left to right, and all six are empty.
10. The block pocket is empty.

## Workspace layout

- **Parts tray:** against the left edge (**Config L**), at the back-center behind the build block (**Config M**), or against the right edge behind the row (**Config R**), within reach of the arm on its side. It holds the bolt holder, the washer dish, and the plate row. Nothing is ever set down on it.
- **Build block:** fixed in the middle of the tabletop, reachable by both arms. Every stack is built here, one at a time.
- **Nut dish:** beside the build block on the supply side, within reach of the supply-side arm (right in Config L and M, left in Config R).
- **Wrench rest:** beside the nut dish on the supply side, within reach of the same arm.
- **Row positions:** six marked spots along the front of the tabletop, running from the middle to the right, numbered 1 to 6 left to right, within reach of the right arm.

Each arm owns the side it sits on. In Config L and M the left arm works the parts tray and steadies the plate, and the right arm works the nuts, the wrench, and the row. In Config R the right arm works the parts tray and steadies the plate, the left arm works the nuts and the wrench, and the right arm still works the row. Both arms reach the build block.

## Vocabulary

- **Arm assignment:** the **tray-side gripper** (left in Config L and M, right in Config R) owns the parts tray, every part staged there, and steadying the plate on the block. The **supply-side gripper** (the other one) owns the nut dish and the wrench. The **right gripper** owns the row positions in every config. Nothing is passed hand to hand.
- **Start zone:** where the parts tray sits at the start of the episode — against the left edge (**Config L**), at the back-center behind the build block (**Config M**), or against the right edge behind the row (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Supply side:** the side opposite the parts tray, where the nut dish and the wrench rest sit — the right in Config L and M, the left in Config R.
- **Build block:** the fixed block in the middle with one pocket. The pocket holds a bolt head down so the bolt cannot turn while the nut is tightened. The block is never gripped, steadied, lifted, or slid.

- **Stack order:** stacks are built one at a time and numbered 1 to 6 in build order. Stack N goes to row position N. No position is skipped and no stack is revisited.
- **Threads up:** the bolt stands in the pocket with its head down inside the pocket and its threaded end pointing straight up.
- **Seated flat:** the washer or the plate lies flat on what is under it, with the bolt through its hole and no visible tilt or gap.
- **Clockwise:** as seen looking straight down at the bolt from above. Tightening is clockwise, loosening is counterclockwise.
- **Quarter turn cycle:** close on the part, turn a quarter turn clockwise, open, bring the gripper back to where it started without touching the part, and close again. Every turn in this SOP is made this way, and no single motion turns further than a quarter turn.
- **Hand tight:** the nut has run down the thread, come to rest flat on the plate, and no longer moves on a quarter turn cycle with the gripper alone.
- **Snug:** hand tight plus two more quarter turns with the wrench, a half turn in all. Nothing is tightened past snug.
- **Tool home:** the wrench rest. The wrench lies there whenever it is not in a gripper, and it goes back after every stack.
- **Release point:** the only places a gripper opens are the block pocket, over the bolt at the block, the row position being filled, and the wrench rest. A gripper never opens over the tray, over a dish, or over open tabletop.
- **Stable placement:** the part or the finished stack stands on its own when the gripper opens, without rocking, leaning, or rolling.
- **Dropped part:** any bolt, washer, plate, or nut that leaves a gripper before its release point. Leave it where it lands and take a fresh part from the tray.

## Steps

Steps 1 to 5 build one stack. Run them in full, six times, in stack order.

Steps 1 to 4 depend on the config: the **tray-side gripper** (left in Config L and M, right in Config R) takes every bolt, washer, and plate and holds the plate, and the **supply-side gripper** takes every nut and the wrench. Step 5 is the same in all three: the **right gripper** carries every finished stack to the row. Every other line is the same in all configs.

### Step 1: Stand a bolt in the build block

**Goal:** one bolt stands in the pocket, head down and threads up, and does not turn in the pocket.

Look where the parts tray is before reaching for the first bolt.

- **IF the tray is against the left edge (Config L):** the **left gripper** takes one bolt from the holder by its shank and lifts it straight up, head hanging below, then carries it level rightward to the build block.
- **IF the tray is at the back-center (Config M):** the **left gripper** takes one bolt from the holder by its shank and lifts it straight up, head hanging below, then carries it level forward to the build block.
- **IF the tray is against the right edge (Config R):** the **right gripper** takes one bolt from the holder by its shank and lifts it straight up, head hanging below, then carries it level leftward to the build block.

Then, in all three:

- Lower the head into the pocket until the bolt stops, and open the gripper.

**Check:** the bolt stands with its head fully in the pocket and its threaded end straight up. If it leans, sits proud of the pocket, or turns freely, seat it again before anything goes on it.

## Step 2: Thread the washer and the plate onto the bolt

**Goal:** the washer sits flat over the bolt head and the plate sits flat on the washer, with the bolt up through both holes.

### 2.1 The washer

- **IF Config L or M:** the **left gripper** closes on the outer edge of one washer in the dish and lifts it straight up, keeping it flat. **IF Config R:** the **right gripper** does the same.
- Carry it level over the bolt, line its hole up with the threaded end, and lower it down the bolt until it rests flat.
- Open the gripper and lift clear.

### 2.2 The plate

- **IF Config L or M:** the **left gripper** closes on two opposite edges of one plate and lifts it straight up, keeping it flat. **IF Config R:** the **right gripper** does the same.
- Carry it level over the bolt, line its hole up with the threaded end, and lower it down the bolt until it rests flat on the washer.
- Do not release the plate. The **tray-side gripper** holds it flat until the nut is snug.

**Check:** the washer and the plate both lie flat with the bolt through their holes, no gap under either, and thread still shows above the plate. If either rides on the thread or sits crooked, lift it off and lower it again before the nut goes on.

## Step 3: Spin the nut down by hand

**Goal:** the nut is hand tight on the plate and the stack is straight.

- **IF Config L or M:** the **right gripper** closes on two opposite flats of one nut in the dish and lifts it straight up, keeping it flat. **IF Config R:** the **left gripper** does the same.
- Carry it level over the bolt, line its hole up with the threaded end, and lower it until it rests on the thread.
- Run a **quarter turn cycle**: turn a quarter turn clockwise, open, bring the gripper back without touching the nut, and close on the flats again.
- Repeat quarter turn cycles until the nut comes to rest flat on the plate and stops moving. That is hand tight.
- Open the gripper and lift clear. The tray-side gripper stays on the plate.

If the nut goes on crooked, binds, or will not run down on the first quarter turn, back it off counterclockwise, lift it off the bolt, and start it again. Never force a nut that is not running down.

**Check:** the nut sits flat and square on the plate, the stack is straight, and thread still shows above the nut.

#### Step 4: Snug the nut with the wrench

**Goal:** the nut is snug, the tray-side gripper is off the plate, and the wrench is back in its rest.

- **IF Config L or M:** the **right gripper** takes the wrench from its rest by the handle and lifts it straight up, keeping it flat. **IF Config R:** the **left gripper** does the same.
- Carry it level over the nut and lower the ring down onto the nut until it sits on the flats. The bolt passes up through the ring.
- Swing the handle a quarter turn clockwise around the nut, keeping the ring seated on the flats.
- Lift the wrench straight up off the nut, bring it back around, and lower it onto the flats again.
- Make **two quarter turns in all** past hand tight, then stop. That is snug.
- Carry the wrench back to its rest, lower it until it lies flat, and release.
- The **tray-side gripper** releases the plate and lifts clear.

Stop at snug. Do not add a third turn, do not pull the wrench further than a quarter turn in one swing, and never use the wrench to run a loose nut down the thread.

**Check:** the nut is snug against the plate, thread still shows above it, the nut flats are not rounded or marked, and the wrench lies flat in its rest.

#### Step 5: Move the finished stack to the row

**Goal:** the finished stack stands in its row position and the block pocket is empty.

- The **right gripper** closes on two opposite edges of the plate and lifts the whole stack straight up out of the pocket, keeping it level. It comes out as one piece, bolt head at the bottom and nut on top.

- Carry it level to the row position for this stack and lower it until the bolt head stands on the tabletop.
- Open the gripper only once the stack stands on its own, then lift clear.

If a part slides on the bolt as the stack lifts, lower it back into the pocket and add one more quarter turn with the wrench before lifting again.

**Check:** the stack stands upright in its position, bolt head down and nut up, not leaning on the stack beside it, and the block pocket is empty.

## **Step 6: Build the remaining five stacks**

**Goal:** six finished stacks stand in row positions 1 to 6.

- Repeat Steps 1 to 5 for stacks 2 through 6, filling row positions 2 through 6 in order.
- Finish one stack and move it to the row before taking the next bolt.
- The wrench goes back to its rest after every stack.

**Check after each stack:** the row holds one more stack than before, in the next position from the left, and the block pocket is empty.

## **Step 7: End the episode**

- Confirm six stacks stand in row positions 1 to 6, each with its washer and plate flat and its nut snug.
- Confirm the bolt holder, the washer dish, the plate row, the nut dish, and the block pocket are all empty.
- Confirm the wrench lies flat in its rest.
- Return both arms home with grippers open, then stop recording.

## **After the episode: reset the workspace**

This reset is not recorded.

1. Take each stack apart: back the nut off counterclockwise, lift the plate and the washer off the bolt, and separate the four parts.
2. Inspect every part. Replace any bolt with damaged threads, any nut with rounded flats, and any plate or washer that is bent or has a burred hole.
3. Set the parts tray for the next episode's config — against the left edge (Config L), at the back-center behind the build block (Config M), or against the right edge behind the row (Config R) — and stand six bolts head down in the holder, threaded ends up, one per hole.
4. Lay six washers flat in the washer dish and six plates flat in the plate row, none stacked or touching.

5. Set the nut dish and the wrench rest on the supply side for that config, and lay six nuts flat in the nut dish, holes facing up, none stacked or touching.
6. Lay the wrench flat in its rest.
7. Clear every loose part off the tabletop, including anything dropped during the episode, and leave the six row positions empty.
8. Check that the build block is still fixed and its pocket is empty.
9. Run both Setup checklists again.

## SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

### How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

### Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

### Violations

**Note on the start position:** the violations below were written for Config L (parts tray against the left edge, nut dish and wrench rest on the right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

#### Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the parts tray, the nut dish, or the wrench rest is not in its place for the config; a gripper reaches across the table for a bolt, a washer, a plate, a nut, or the wrench; the supply-side gripper steadies the plate; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the supply side, and the same-side rule (the tray-side gripper takes every bolt, washer, and plate and holds the plate; the supply-side gripper takes every nut and the wrench; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the parts tray is before the first reach, then follow that config's IF lines through Steps 1 to 4.

#### Violation: Parts added in the wrong order

- **Visible cue:** the plate goes onto the bolt before the washer, the nut is started before the plate is on, or a washer or plate is added after the nut has been started.
- **SOP rule broken:** Steps 2 and 3, washer, then plate, then nut, on every stack.
- **Coaching note:** washer, plate, nut. Same order every time.

#### **Violation: Wrong stack or row order**

- **Visible cue:** a finished stack goes to a position out of order, a position is skipped, two stacks share one position, or the build goes back to an earlier position.
- **SOP rule broken:** Step 5, stack N goes to row position N, left to right.
- **Coaching note:** fill the row left to right, one stack per position.

#### **Violation: Bolt not seated in the pocket**

- **Visible cue:** parts go onto a bolt that leans, sits proud of the pocket, or turns freely in the pocket when the nut is spun.
- **SOP rule broken:** Step 1 (the head sits fully in the pocket and the bolt does not turn before anything goes on it).
- **Coaching note:** seat the head and check it will not turn before you add the washer.

#### **Violation: Part not seated flat before the next part**

- **Visible cue:** the washer or the plate is left tilted, riding on the thread, or bridging the pocket, and the next part goes on anyway.
- **SOP rule broken:** Step 2, lift it off and lower it again before the nut goes on.
- **Coaching note:** look under it before you add the next part.

#### **Violation: Turned further than a quarter turn in one motion**

- **Visible cue:** the gripper turns the nut, or the wrench swings, through more than a quarter turn without opening or lifting off, or the forearm runs into its limit and the part stops turning under load.
- **SOP rule broken:** Steps 3 and 4, no turn goes past a quarter turn in one motion.
- **Coaching note:** quarter turn, let go, come back, close again.

#### **Violation: Nut forced when it is not running down**

- **Visible cue:** the nut sits crooked on the thread, binds, or stops moving down, and a gripper keeps turning it.
- **SOP rule broken:** Step 3 (back it off, lift it off the bolt, and start it again; never force a nut that is not running down).
- **Coaching note:** if it is not running down, take it off and start it again.

#### **Violation: Nut not hand tight before the wrench**

- **Visible cue:** the wrench comes onto a nut that is still standing up the thread, and the wrench is used to run it down to the plate.
- **SOP rule broken:** Step 4, never use the wrench to run a loose nut down the thread.
- **Coaching note:** the gripper runs it down. The wrench only snugs it.

#### **Violation: Wrong snug amount**

- **Visible cue:** the wrench makes fewer or more than two quarter turns past hand tight, or the wrench keeps pulling after the second turn.
- **SOP rule broken:** Step 4, two quarter turns past hand tight, then stop.
- **Coaching note:** count the two turns out loud and stop.

#### **Violation: Wrench not seated on the flats**

- **Visible cue:** the wrench is swung while its ring sits part way on the nut, catches the plate, or slips off the nut during the swing.
- **SOP rule broken:** Step 4 (lower the ring onto the nut until it sits on the flats before swinging).
- **Coaching note:** seat it fully, then pull.

#### **Violation: Plate not steadied**

- **Visible cue:** the left gripper comes off the plate before the nut is snug, and the plate turns with the nut or ends up crooked on the washer.
- **SOP rule broken:** Steps 2.2, 3, and 4 (the left gripper holds the plate flat until the nut is snug).
- **Coaching note:** the left hand stays on the plate until the wrench is back in its rest.

#### **Violation: Build block gripped, steadied, or moved**

- **Visible cue:** a gripper closes on the block or presses against it to hold it still, or the block is pushed out of place during the episode.
- **SOP rule broken:** Steps 1 to 6, the build block is never gripped, steadied, or moved.
- **Coaching note:** the block is fixed and holds itself. If it is moving, stop and have it fixed again.

#### **Violation: Wrench left outside its rest**

- **Visible cue:** the wrench is set down on the tabletop, on the tray, on the block, or in the row instead of going back to its rest, or it is still in the gripper when the next bolt is taken.
- **SOP rule broken:** Step 4, the wrench goes back to its rest after every stack.
- **Coaching note:** the wrench has one home. It goes there every time.

#### **Violation: Wrong gripper on a part or a zone**

- **Visible cue:** the right gripper picks a bolt, a washer, or a plate off the tray, or the left gripper takes a nut, takes the wrench, or reaches a row position.
- **SOP rule broken:** Steps 1 to 6, each gripper works only its own parts and zones.
- **Coaching note:** the left arm loads the tray parts and holds the plate. The right arm does the nut, the wrench, and the row.

#### **Violation: Part passed hand to hand**

- **Visible cue:** a bolt, washer, plate, nut, or the wrench is transferred straight from one gripper to the other in mid air.
- **SOP rule broken:** Steps 1 to 6, nothing is passed hand to hand.
- **Coaching note:** nothing changes hands in the air.

#### **Violation: Gripper opened over the wrong place**

- **Visible cue:** a gripper opens over the tray, over a dish, or over open tabletop, or a part is released anywhere other than the block, its row position, or the wrench rest.
- **SOP rule broken:** Steps 1 to 6, a gripper opens only over the spot the part is going to.
- **Coaching note:** know where it lands before you open the gripper.

#### **Violation: Dropped part collected mid episode**

- **Visible cue:** an arm stops to pick up a bolt, washer, plate, or nut that has landed on the tabletop, instead of taking a fresh part from the tray and carrying on.
- **SOP rule broken:** Steps 1 to 6, a dropped part is left where it falls.
- **Coaching note:** leave it. Dropped parts are cleared in the reset, not on camera.

#### **Violation: Stack lifted before it is snug**

- **Visible cue:** the stack is lifted out of the pocket while the nut is still loose, and the plate or the washer slides on the bolt or the stack comes apart.
- **SOP rule broken:** Steps 4 and 5 (snug the nut, then lift the finished stack by the plate).
- **Coaching note:** snug it first. A finished stack lifts as one piece.

#### **Violation: Stack released before a stable placement**

- **Visible cue:** the gripper opens while the stack is still leaning, tipping, or held above the tabletop, or the stack falls over after the gripper opens.
- **SOP rule broken:** Step 5, open only once the stack stands on its own.
- **Coaching note:** let it stand on its own before you open.

#### **Violation: Item dropped, collided, or knocked over**

- **Visible cue:** a bolt is knocked out of the holder, a dish or the tray is dragged or tipped, a finished stack in the row is struck out of its position, or an arm hits the build block hard

enough to move it.

- **SOP rule broken:** Steps 1 to 5 (carry every part level and clear of what it passes over, and release only after a stable placement).
- **Coaching note:** check the path and the landing spot before you move.

### **Violation: Wrong episode ending**

- **Visible cue:** the episode ends with fewer than six stacks in the row, a part still in the tray or a dish, a stack still on the block, the wrench out of its rest, an arm away from home or a gripper closed.
- **SOP rule broken:** Step 7 (confirm the six stacks and the empty zones, return both arms home, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

### **Failures that are not violations**

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective part:** a bolt with damaged threads, a nut that will not start on any bolt, or a plate or washer whose hole is burred so it will not pass over the bolt. Replace it before the next episode.
- **Build block fault:** the block works loose, or its pocket no longer holds the bolt head from turning, so the nut cannot be tightened. Fix it again before the next episode.
- **Wrench or rest fault:** the ring does not fit the nut flats, the wrench comes loose in the gripper, or the rest does not hold the wrench flat.
- **A dish, the tray, or the bolt holder shifts out of its zone** during the episode.

### **Annotation subtasks (from SOP)**

1. Stand a bolt in the build block
2. Thread a washer down onto the bolt
3. Lower a plate onto the bolt
4. Spin a nut down to hand tight
5. Take the wrench from its rest
6. Snug the nut with the wrench
7. Return the wrench to its rest
8. Move a finished stack to its row position

9. Return both arms home and end the episode